Theory-Based User Modeling with Grounded Transformers

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Abstract. The user model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing approaches, which represent the learner's state of mastery across a set of skills or knowledge components. However, these approaches are not anchored to theoretical frameworks derived from learning theories, which limits their utility when it comes to interpreting the evolution of user profiles and learning trajectories. To overcome these limitations, we propose a new approach based on Grounded Transformers, that allow to augment the user model with information grounded in learning theories.

Keywords: deep learning · theory-based interpretability · grounded transformers · user modeling · learning theories · intelligent tutoring systems

1 Introduction

Intelligent Tutoring Systems (ITS) rely on a student model to represent the learner's evolving knowledge state and to drive adaptive instructional decisions. In most deployed systems, this model is operationalised as a set of mastery estimates over a predefined collection of skills or knowledge components, updated incrementally as the student interacts with learning materials. Knowledge Tracing (KT) approaches, ranging from Bayesian Knowledge Tracing (BKT) [1] to deep learning architectures such as DKT [2] and attention-based models [3], have substantially advanced the accuracy of these estimates. However, predictive accuracy alone is insufficient for purposeful adaptive instruction: a student model that cannot articulate why a learner is struggling, or that cannot provide theory-grounded predictions of future performance, offers limited diagnostic value to educators and tutoring systems alike.

Two fundamental limitations characterise current KT models. First, high-performance neural approaches are largely opaque: their internal representations do not correspond to pedagogically recognisable constructs, making it difficult for practitioners to justify or act on model outputs. Second, interpretable models such as standard BKT operate at the population level, estimating a single parameter set for all students on a given skill [4], and therefore cannot capture the individual learning patterns that determine which instructional strategy will be most effective for a given learner at a given moment.

Recent work on Grounded Transformers [5] addresses both limitations by constraining Transformer latent representations to align with the semantic axes of reference models like BKT through *representational grounding*. This yields student-specific, longitudinally updated estimates of initial mastery (p_{L_0}) and learning rate (p_{T}), derived from each student's full interaction history rather than from population averages. In the present work we investigate how these grounded parameters can augment the ITS student model in ways that support richer and more intentional instructional strategies.

We propose an *augmented student model* that integrates Grounded Transformer parameters into the ITS user model and show how doing so enables context-aware, *situation-based instruction*. In [5], they evaluate gTransformer on five benchmark datasets and show that interpretable predictions derived from grounded parameters consistently outperform classical BKT by an average AUC gain of 19.9%, at a grounding cost of only 3.9% relative to black-box deep learning architectures. They further confirm, through structural, semantic, and functional alignment analyses, that the model's representations genuinely encode BKT constructs as primary organising principles rather than as surface correlates. These properties make the augmented student model that we propose a viable foundation for a more adaptive, human-centred ITS.

The validation of our proposal is guided by the following research question: how can augmenting a user model with Grounded Transformer parameters support the design of context-aware, situation-based instructional strategies that are both grounded in established learning theories and mindful of the evolving needs of individual learners?

The remainder of the paper is organised as follows. Section 2 formalises the proposed theory-based augmented user model. Section 3 describes the experimental methodology and evaluation framework. Section 4 presents the results across the three ITS components. Section 5 details the experimental setup and baselines. Section 6 demonstrates context-aware personalisation through learning situation routing. Section 7 discusses the broader implications, and Section 8 concludes.

1.1 Intelligent Tutoring Systems and User Models

Intelligent Tutoring Systems are typically structured around three interacting components:

- *Domain model*: This component encodes the knowledge to be taught, typically structured as a set of Knowledge Components (KCs) or skills. It provides the base through which student performance is assessed and instructional content is organised.
- *User model*: The user model tracks the learner's evolving state by maintaining a representation of their mastery across the domain model.
- *Tutorial model*: This component mediates instructional decisions by selecting the next activity, intervention, or level of support based on the current state of the user model. The goal is to provide a personalised learning experience that optimises the student's trajectory.

1.2 Learning Theories and Bayesian Knowledge Tracing

We select BKT [1] as our reference theoretical framework. BKT models the learning process as a Hidden Markov process governed by four interpretable parameters: Initial Mastery ($P(\mathrm{L}_0)$), the probability that a student knows a concept prior to practice; Learn Rate ($P(\mathrm{T})$), the probability of transitioning from the unlearned to the learned state; Guess ($P(\mathrm{G})$), the probability of a correct response despite lack of mastery; and Slip ($P(\mathrm{S})$), the probability of an incorrect response despite mastery. These parameters enable a recursive, causally transparent estimation of student mastery that is directly interpretable by educators.

Standard BKT suffers from a significant limitation: it operates at the population level, estimating a single parameter set for all students on a given skill [4].

2 Grounded Transformers and Augmented User Models

2.1 Proposed Architecture

We propose an *augmented user model* that integrates the three standard ITS components with a Grounded Transformer engine. The *domain model* defines the structured knowledge components to be mastered. The *user model* tracks student mastery across these components and is augmented with student-specific parameters $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ inferred by gTransformer [5] from the student's full interaction history. These parameters encapsulate the student's longitudinal learning context, allowing the user model to differentiate learners who show similar correctness patterns but differ in prior knowledge or learning rate. The outputs of this augmented user model feed: 1) the *tutorial model*, which selects situation-based instructional strategies; and 2) a *diagnostic roster module*, which visualises student trajectories in $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ space to support educator oversight.

2.2 gTransformer

The augmented user model is powered by gTransformer [5], a hybrid architecture that implements *representational grounding*: a methodology that anchors Transformer latent representations to the semantic constructs of BKT, bridging deep learning expressiveness with pedagogical interpretability. Figure 1 summarises the architecture.

As illustrated in Figure 1, the architecture operates through two parallel tracks. The *standard track* encodes student interactions (q_t, c_t, r_t) via difficulty-aware embeddings and processes them through a stacked attention-based encoder–decoder, producing a context vector z_t that summarises the student's current learning state. A multi-layer perceptron maps z_t directly to a correctness prediction $\hat{y}_t$, optimising for predictive accuracy without theoretical constraints.

The *grounded track* implements representational grounding by composing student-specific BKT parameters in logit space. For each knowledge component, logit-transformed population-level BKT priors serve as theory base values; contextual adjustments are obtained by projecting z_t onto learnable concept-specific semantic axes for initial mastery and learning rate. The resulting grounded parameters $p_{\mathrm{L}_0,t} = \sigma(\ell_{\mathrm{L}_0} + z_t \cdot k_c)$ and $p_{\mathrm{T},t} = \sigma(\ell_{\mathrm{T}} + z_t \cdot v_c)$ are then passed through BKT update logic to produce interpretable predictions $\hat{y}_{\mathrm{ref},t}$. Training is governed by a multi-objective loss balancing supervised accuracy, interpretable prediction quality, and global latent-space alignment through diagnostic probing. Full architectural details are provided in [5].

3 Methodology

To address RQ1, we evaluate *gBKT* by comparing its predictive performance against traditional population-level BKT. Unlike standard BKT which uses constant parameters for all students on a given skill, gBKT leverages gTransformer to infer context-aware parameters from historical interaction sequences. gTransformer uses a Transformer-based encoder-decoder architecture where latent representations z_t are projected onto BKT semantic axes through representational grounding. The model is trained on benchmark datasets such as AS2009, with the grounded parameters serving both to constrain predictions to pedagogical logic and to augment the student profile.

To address RQ2, we define four *learning situations* based on the estimated $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$: (1) *foundational learners* (low L_0, low T), (2) *emerging learners* (low L_0, high T), (3) *consolidating learners* (high L_0, low T), and (4) *advancing learners* (high L_0, high T). These situations allow the tutorial model to adapt strategies: foundational learners might receive targeted scaffolding, while advancing learners receive accelerated content. Since parameters are dynamic, students can transition between learning situations as more evidence is gathered. To illustrate this, we generate 3D roster plots showing the longitudinal trajectory of representative students across these learning situations.

4 Results and Discussion

Comparison of predictive performance (Table 1) addresses RQ1, showing that gBKT significantly outperforms standard BKT, with an average AUC gain of 19.9% ($\mathcal{I}_{\mathrm{gain}}$). This validates that context-aware parameters derived from grounded Transformers accurately capture the nuances of individual learning journeys while remaining within the interpretable framework of BKT.

Figure 2 addresses RQ2, showing trajectory plots for a representative student from each of the four learning situations. Each panel maps consecutive sampled interactions in (p_T, p_{L_0}) space, with numbered points and directional arrows tracing the temporal evolution of the student's parameter profile across the full interaction sequence. Coloured background regions delimit the four learning situations; a point may cross boundaries, indicating a genuine shift in the student's learning situation over time. By reading these trajectories, educators gain theory-grounded insights into whether a student is disengaging (decreasing p_{T}) or gaining confidence (increasing p_{L_0}). This dynamic view grounds deep learning state into pedagogically recognisable constructs, enabling more mindful instructional support and satisfying the accountability requirements for high-risk educational AI.

The parameter orbits of representative students across the four learning situations are characterised in Figures 3 and 4. A key property of gTransformer's augmented user model is that the learning situation label assigned to each student is not a static category frozen at enrolment: it is a summary of the student's longitudinal tendency, continuously updated as new interactions arrive. The covariance ellipses in Figure 3 make this explicit: each ellipse visualises the full spread of $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ values that a student's profile traverses across all their interactions, rather than a single point estimate.

For *foundational learners*, the ellipse is compact and centred at low mastery and low learning rate, indicating limited contextual variation: across skills and interactions, the model consistently reads this student as struggling to progress. This persistence is itself instructionally informative: it is not that no learning is occurring, but that the rate is systematically insufficient given the current instructional conditions. The tutorial model should not wait for cumulative failure before acting; the ellipse signals a structural mismatch between task demands and student preparation that calls for immediate scaffolding, worked examples, and reduced problem complexity. Crucially, this diagnosis can be revised: if the student responds to the intervention and p_{L_0} or p_{T} begins to rise, the orbit will shift and the learning situation label will update accordingly.

For *consolidating learners*, the ellipse is compact and located at high mastery and low learning rate. The small spread indicates that the student has reached a stable region of the parameter space with little contextual variation: prior knowledge is high but the per-interaction learning rate has converged to near zero. This pattern is not a failure; it reflects a student who has consolidated existing knowledge, but it does signal stagnation. The tutorial model should interpret this as a cue to introduce novel, cross-domain challenge or to advance the student to a higher difficulty tier, as continued practice at the current level is unlikely to produce further gains. If challenge is appropriately increased and p_{T} recovers, the ellipse will redistribute and the tutorial model will update its routing.

For *emerging learners*, the ellipse is the most spatially extended of all four situations, oriented along both the mastery and learning-rate axes. This broad orbit reflects a student whose parameters vary substantially across skills and interaction contexts: at some skills the student is already proficient; at others, they are still building foundations, but with a high and increasing learning rate. This heterogeneity is precisely what a static label would erase. The tutorial model should exploit it by differentiating at the skill level, providing scaffolding on weaker knowledge components while allowing acceleration on stronger ones, rather than applying a uniform strategy to the whole student.

For *advancing learners*, the ellipse is moderately extended and centred at high mastery with a moderate learning rate. The spread reflects a student who engages responsively across varied skill demands while maintaining consistently high prior knowledge. The tutorial model can safely route this student to accelerated content and reduced practice redundancy; the ellipse shape confirms that the high-mastery profile is robust across contexts and not an artefact of easy items.

Figure 4 shows plots for the same four representatives. Each panel plots the value of a parameter at interaction $t+1$ against its value at interaction t, separately for p_{L_0} (left sub-panel) and p_{T} (right sub-panel), with points coloured from dark (early interactions) to bright (late interactions) using a sequential palette. The diagonal represents a fixed point: points on the diagonal indicate no change between consecutive interactions. Deviations above the diagonal signal an increasing parameter; deviations below signal a decrease. Unlike population-level BKT, which produces a single pair of constant parameter values for all students on a given skill, each return map encodes the unique temporal trajectory of an individual student, a view that is both dynamic and context-aware, and one that evolves measurably as the student progresses.

For *foundational learners*, the p_{L_0} return map reveals a broad scatter around the diagonal with a systematic downward drift from early interactions (mean 0.61) to late interactions (mean 0.48): the model progressively revises its mastery estimate downward as evidence of insufficient learning accumulates. The p_{T} panel is tightly concentrated near zero throughout, exposing a stable low-rate attractor that a snapshot profile would reduce to a single low number. The trajectory, however, shows that the attractor consolidates over time; the orbit is converging, not oscillating randomly. This temporal structure informs instructional timing: an intervention delivered before the orbit locks into the low attractor is far more likely to dislodge the student from this trajectory than one applied after convergence. The tutorial model, monitoring this return map in real time, can act early and revise the learning-situation assignment as soon as p_{T} or p_{L_0} shows a sustained upward deviation.

For *consolidating learners*, the p_{L_0} return map undergoes a visible temporal transformation: early points span the mid range, while late points consolidate near the diagonal at high mastery (0.80), forming a tight cluster. Simultaneously, p_{T} declines from 0.077 to 0.050, converging toward a low-rate fixed point. The orbit has not always been in this state: the student arrived at the plateau through accumulated practice, and the return map preserves that history rather than collapsing it into a single label. A static model would see only high mastery and low learning rate; the dynamic view reveals that this student is transitioning from active learning to consolidation. The tutorial model can exploit this information to introduce transferable challenge or cross-domain tasks that reactivate the learning trajectory before the orbit fully stabilises, rather than treating the plateau as an irreversible state.

For *emerging learners*, both p_{L_0} ($0.52 \rightarrow 0.67$) and p_{T} ($0.12 \rightarrow 0.20$) increase over time, with points distributed above the diagonal in both panels. This is the return map of a student whose orbit has not yet converged: the system is continuously revising its estimates upward as evidence of genuine learning accumulates interaction by interaction. A fixed label would misrepresent this student either as a weak learner (based on early interactions) or as a competent one (based on late interactions). The dynamic profile captures both states and the transition between them, enabling the tutorial model to progressively accelerate pacing and increase challenge level in real time, matched to the observed trajectory rather than to a predetermined category.

For *advancing learners*, the p_{L_0} return map shows early stabilisation: mastery is high from the first interactions (mean 0.74) and the orbit clusters around the diagonal with negligible net movement. The p_T panel declines moderately (0.15 $\rightarrow$ 0.11), indicating that the model assigns progressively smaller marginal learning rate estimates as opportunities for further gain diminish. The instructional implication is not derived from a presumed initial ability but from observed evidence: continued practice at the current difficulty level produces diminishing returns, and the tutorial model can route this student to more demanding content immediately, grounded in the trajectory rather than in a prior assignment.

These results underscore the utility of theory-grounded user modeling for ECTEL 2026's theme of Mindful TEL. By shaping learning technologies with intention, we ensure that deep learning's predictive power does not come at the cost of pedagogical transparency. The augmented user model provides the "diagnostic handles" required for educators to exercise agency within AI-mediated environments, transforming the "black box" into a collaborative cognitive engine that supports, rather than replaces, human expertise.

5 Experimental Setup

We evaluate gTransformer on five benchmark datasets with explicit knowledge component mappings: ASSISTments 2009 (AS2009, 4,217 students, 123 skills), ASSISTments 2015 (AS2015, 19,917 students, 100 skills), Algebra 2005 (AL2005, 574 students, 112 skills), Bridge to Algebra 2006 (BDG2006, 1,146 students, 493 skills), and NeurIPS 2020 Education Challenge (NIPS34, 4,918 students, 57 skills). We follow a standard five-fold cross-validation with an 80/20 train/test split, measuring performance via AUC using the question-level mean late-fusion protocol [6]. Baselines include DKT [2], DKVMN [7], SAKT [8], SAINT [9], AKT [3], and ATKT [10].

5.1 User Model: Theory-Grounded Knowledge Tracing

To quantify gTransformer's performance as a user model and the cost of pedagogical grounding, we define three metrics. Let p_{sup1} be ablated (non-grounded) supervised predictions, p_{sup2} grounded supervised predictions, p_{ref} interpretable BKT-based predictions, and p_{bkt} classical BKT predictions:

$$\mathcal{I}_{\text{drop}} = p_{\text{sup1}} - p_{\text{sup2}} \quad \text{(cost of grounding on neural capacity)} \tag{1}$$
$$\mathcal{I}_{\text{cost}} = p_{\text{sup2}} - p_{\text{ref}} \quad \text{(cost of switching to interpretable logic)} \tag{2}$$
$$\mathcal{I}_{\text{gain}} = p_{\text{ref}} - p_{\text{bkt}} \quad \text{(advantage over population-level BKT)} \tag{3}$$

Table 1 shows the comparative performance of gTransformer against baselines in the ablated (non-grounded) configuration. gTransformer achieves the best

Table 1: **Predictive performance (AUC) comparison.** Bold indicates best performance per dataset.

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
AKT	0.7825	0.7081	**0.8306**	**0.8208**	**0.8033**
ATKT	0.7715	**0.7810**	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873
gTransformer	**0.7831**	0.7078	0.8240	0.8148	0.8006

Table 2: **AUC–interpretability trade-offs.** $\mathcal{I}_{\text{drop}}$ = cost of grounding; $\mathcal{I}_{\text{cost}}$ = interpretability cost; $\mathcal{I}_{\text{gain}}$ = gain over BKT.

Dataset	p_{sup1}	p_{sup2}	p_{ref}	p_{bkt}	$\mathcal{I}_{\text{drop}}$	$\mathcal{I}_{\text{cost}}$	$\mathcal{I}_{\text{gain}}$
AS2009	0.7831	0.7814	0.7436	0.6097	0.2%	4.8%	22.0%
AS2015	0.7078	0.7070	0.6940	—	0.1%	1.8%	—
AL2005	0.8240	0.8237	0.7800	0.7215	0.0%	5.3%	8.1%
BDG2006	0.8148	0.8107	0.7810	0.6756	0.5%	3.6%	15.6%
NIPS34	0.8006	0.7987	0.7666	0.5729	0.2%	4.0%	33.8%
Mean	0.7861	0.7843	0.7530	0.6449	0.2%	3.9%	19.9%

AUC on AS2009, ranks second on AL2005, BDG2006, and NIPS34, and third on AS2015, demonstrating that the architecture is competitive with the state of the art.

Table 2 quantifies the trade-off. The interpretability drop ($\mathcal{I}_{\text{drop}}$) is negligible on average (0.2%), demonstrating that grounding constraints do not compromise the model's expressive capacity. The interpretability cost ($\mathcal{I}_{\text{cost}}$), i.e., the performance reduction when constraining predictions to BKT logic, averages 3.9% across datasets. Against population-level BKT, interpretable gTransformer predictions show a consistent advantage of 19.9% on average. These results establish that pedagogical interpretability can be achieved at a minimal performance cost.

5.2 Domain Model: Validating Theoretical Grounding

To verify that gTransformer's internal representations are genuinely grounded in the domain model (BKT), we implement a triple-validation methodology: (H2.1) Structural Alignment, assessed via diagnostic probing [11,12] measuring the linear recoverability of BKT constructs from hidden states; (H2.2) Semantic Alignment, assessed via Spearman's rank correlation (ρ) between grounded parameters and theoretical priors; and (H2.3) Functional Alignment, assessed via a composite confidence metric quantifying the reliability of interpretable predictions as substitutes for supervised ones. We validate using AS2009 as the reference dataset.

Structural Alignment (H2.1). Using diagnostic probing, we assess whether BKT constructs are linearly recoverable from final hidden states. Initial Mastery achieves fidelity $R^2 = 0.549$ ($r = 0.743$) with selectivity $\Delta R^2 = 0.619$, exceeding the 0.5 threshold that distinguishes genuine structural encoding from spurious correlations. Learn Rate achieves $R^2 = 0.515$ ($\Delta R^2 = 0.570$). These results confirm that BKT constructs are preserved as primary organising principles in the latent space.

Semantic Alignment (H2.2). Initial Mastery shows weaker monotonic alignment with theoretical priors ($\rho = 0.360$, MAE $= 0.168$), reflecting strong student-specific individualisation. Learn Rate demonstrates moderate alignment ($\rho = 0.544$, MAE $= 0.092$), preserving the pedagogical understanding of practice effects. Both parameters remain within pedagogical bounds, confirming that the model neither abandons nor mechanically reproduces theoretical priors.

Functional Alignment (H2.3). A composite confidence metric shows that 89.2% of student–skill pairs achieve medium or high confidence in interpretable predictions, indicating that theory-grounded outputs provide actionable diagnostic value.

5.3 Tutorial Model: Situation-Based Instruction

The theory-grounded parameters produced by gTransformer (student-specific estimates of initial mastery (p_{L_0}) and learn rate (p_{T})) provide the diagnostic handles through which an ITS tutorial model can implement *situation-based instruction*. Rather than applying a uniform instructional strategy, the tutorial model can route each student to a differentiated instructional pathway based on their learning situation as identified by gTransformer. The following section describes how this routing is operationalised through context-aware personalisation.

6 Context-Aware Personalisation

A key contribution of this work is demonstrating how gTransformer enables the ITS tutorial model to move from population-level strategies towards situation-based, context-aware instruction. Rather than assigning students to fixed ability groups, we identify four *learning situations* by partitioning the $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ parameter space along two dimensions (initial mastery and learning rate) on the AS2009 dataset: (1) *foundational learners* (low initial mastery, low learning rate), (2) *emerging learners* (low initial mastery, high learning rate), (3) *consolidating learners* (high initial mastery, low learning rate), and (4) *advancing learners* (high initial mastery, high learning rate). These learning situations are not static categories: because gTransformer continuously updates its parameter estimates as new interactions arrive, a student's assigned situation evolves with their trajectory. Each situation calls for a distinct tutorial strategy: foundational learners require targeted scaffolding; advancing learners are ready for accelerated pacing; intermediate situations benefit from adjusted challenge levels or motivational support.

Figure 5 shows mastery trajectories for students who provide identical response sequences to the same questions but belong to different learning situations. Classical BKT predictions (dotted grey lines) overlap perfectly, reflecting its Markovian constraint: given the same response sequence, BKT produces identical probability estimates regardless of learning history. gTransformer predictions (solid coloured lines) diverge based on the full interaction history, demonstrating the model's capacity to differentiate students not by what they answered, but by how they have learned across their entire trajectory.

This divergence has direct implications for the tutorial model. Each student interaction event can be routed to a different instructional pathway based on the learning situation assigned by gTransformer. A student whose trajectory places them in the foundational learning situation triggers a scaffolding strategy; a student identified as an advancing learner is routed towards accelerated content progression. Crucially, this routing is not static: as new interactions arrive, gTransformer updates its parameter estimates from the full longitudinal context, allowing the tutorial model to dynamically reassign students across situations as their learning trajectory evolves. This event-driven, context-aware routing is precisely what population-level models such as standard BKT cannot support: two students with an identical recent response pattern may receive different instructional interventions if their prior histories place them in different learning situations.

7 Discussion

7.1 Theory-Grounded Instructional Decisions

The results demonstrate that gTransformer addresses the accuracy–interpretability trade-off in KT by providing pedagogically grounded parameters at a minimal performance cost. This is directly relevant to the ECTEL 2026 theme of Mindful TEL: the model embodies the principle that learning technologies should be shaped with intention, ensuring that the cognitive engine of an ITS is anchored in established educational theory rather than opaque data-driven heuristics.

By operationalising BKT parameters (P_{L_0}, P_{T}) derived from deep learning context, gTransformer provides educators with the diagnostic handles through which an ITS can perform mindful instruction. Rather than acting as a black box that issues recommendations without justification, the system grounds its decisions in constructs that practitioners recognise and can validate. This aligns with broader EU imperatives for explainable, accountable AI in education [13], and responds directly to the concerns raised by the Artificial Intelligence Act's designation of educational AI as high-risk.

7.2 Beyond Static Labelling: Context-Aware Adaptation

A central contribution of this work is the shift from population-level labelling to situation-based diagnosis. The four learning situations (foundational, emerging, consolidating, and advancing) are not fixed assignments but dynamic summaries of each student's current longitudinal profile. The results presented in Section 6 demonstrate that gTransformer's context-aware predictions provide a higher-resolution understanding of the learning journey than population-level models. By encoding each student's longitudinal history into updatable parameters, the model acknowledges that students are dynamic learners whose instructional needs evolve continuously.

This enables educators to identify, for instance, students in the consolidating learning situation (high prior knowledge but low learning rate, which may signal disengagement or instructional mismatch) and adapt the pacing accordingly before significant learning gaps emerge. Conversely, students in the advancing situation can be identified for accelerated progression to prevent unnecessary remediation. Crucially, a student who responds positively to an intervention may transition from a foundational learning situation to an emerging one, and the tutorial model will update its routing accordingly. This intentional, responsive design addresses essential concerns about accountability and pedagogical integrity in automated instruction.

7.3 Pedagogical Integrity and Human Agency

The triple-validation methodology (RQ2) establishes that the model's internal representations genuinely internalise pedagogical theory rather than merely correlating with it. Structural alignment confirms BKT constructs as primary organising principles; semantic alignment confirms that grounded parameters retain their pedagogical semantics; and functional alignment quantifies the confidence with which educators can trust interpretable predictions. Together, these results support the claim that gTransformer is not merely a high-performance prediction system, but a theory-grounded diagnostic tool that can serve as the cognitive foundation of responsible, human-centred ITS.

8 Conclusions

This work demonstrates how gTransformer [5], a Grounded Transformer that bridges deep learning performance and pedagogical interpretability through *representational grounding*, can serve as the cognitive foundation of a theory-grounded ITS. We evaluated its capacity to instantiate the three core ITS components. As a user model, interpretable gTransformer predictions consistently surpass classical BKT with an average gain of 19.9% AUC, at a low interpretability cost of 3.9% relative to black-box architectures. As a domain model, gTransformer internalises BKT constructs as its primary organising principle in latent representations, confirmed by structural, semantic, and functional alignment. As a tutorial model foundation, gTransformer enables context-aware, situation-based instruction by capturing longitudinal learning patterns that Markovian models cannot represent.

As a foundation for TEL, gTransformer demonstrates that pedagogically justifiable student modeling is achievable without sacrificing the predictive power needed for effective adaptive instruction. The *representational grounding* methodology is domain-independent and may be extended to other theoretical frameworks beyond BKT or to other scientific fields where established theoretical models coexist with high-volume empirical data. Future research will investigate the integration of guess and slip parameters for finer-grained personalisation and the generalisation of gTransformer to alternative educational theories.

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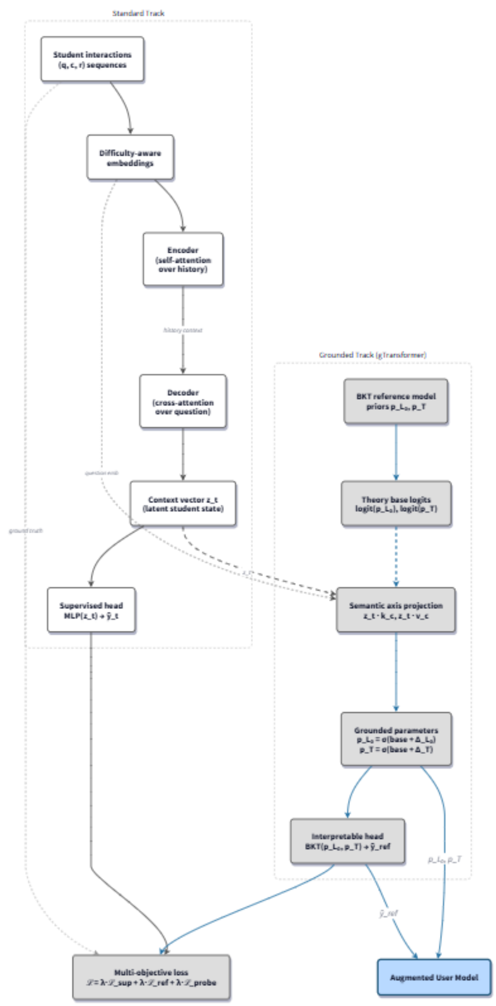
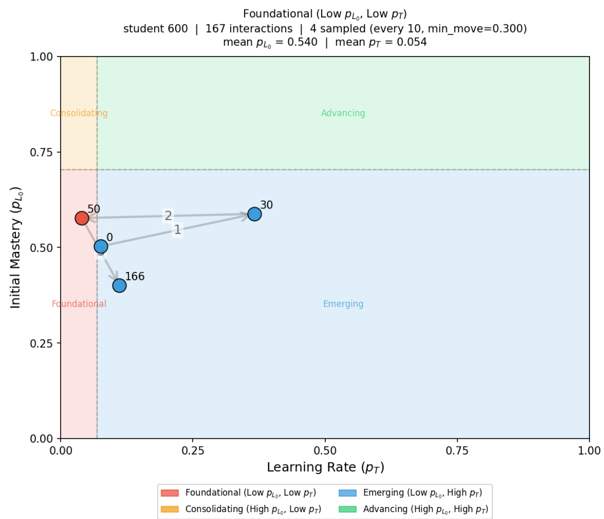
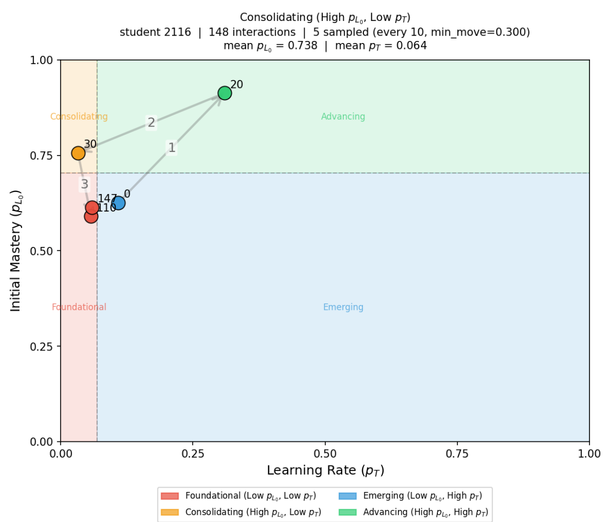


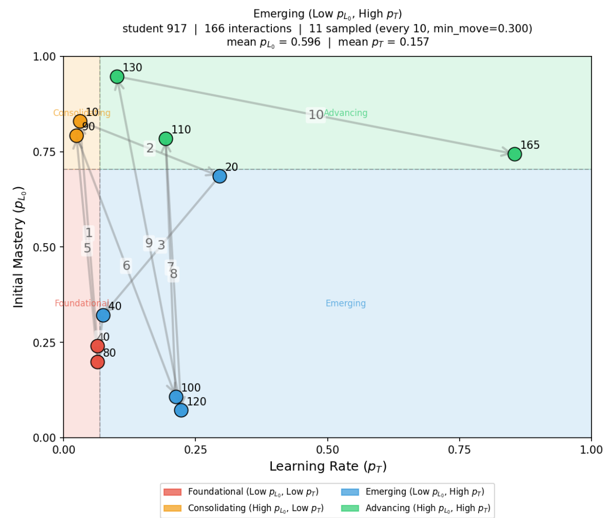
Fig. 1: **gTransformer architecture.** Two processing tracks share a common encoder–decoder backbone. The standard track (white boxes) produces supervised predictions via an MLP head. The grounded track (grey boxes) combines BKT theory base logits with context-vector projections onto learnable semantic axes to produce student-specific parameters (p_{L_0}, p_T), which are passed through BKT update logic to yield theory-transparent predictions. Both heads are jointly optimised via a multi-objective loss. Outputs of the grounded track populate the augmented user model (blue box).



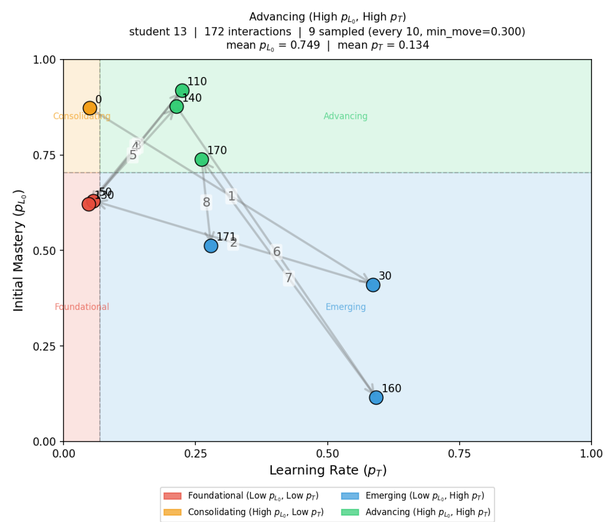
(a) *Foundational* (low p_{L_0}, low p_T): trajectory confined to the lower-left quadrant; limited spread signals a structural mismatch calling for immediate scaffolding.



(b) *Consolidating* (high p_{L_0}, low p_T): compact trajectory anchored at high mastery and near-zero learning rate; stagnation cues the introduction of novel challenge.

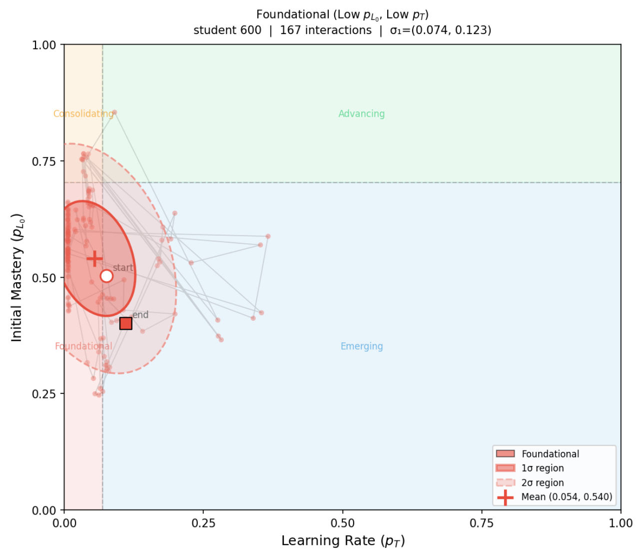


(c) *Emerging* (low p_{L_0}, high p_T): spatially extended trajectory crossing quadrant boundaries; high learning rate signals strong growth potential.

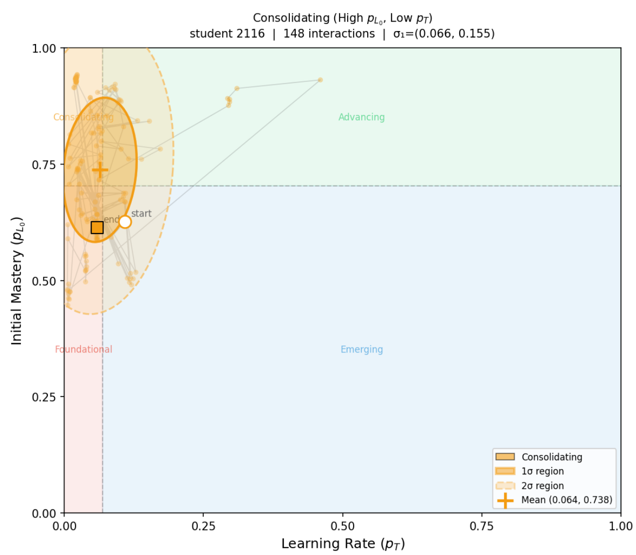


(d) *Advancing* (high p_{L_0}, high p_T): upper-right trajectory with moderate spread; robust mastery across contexts supports routing to accelerated content.

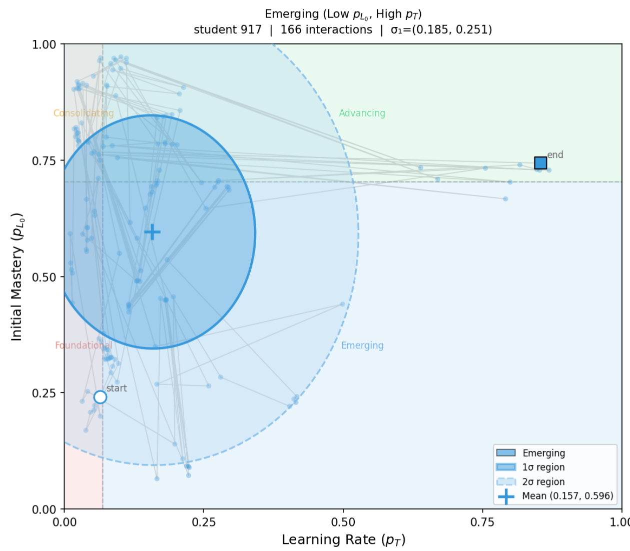
Fig. 2: **Learning situation trajectories.** Each trajectory traces the evolution of a representative student's parameter pair (p_T, p_{L_0}) over successive interactions. A candidate is considered every 10 interactions and plotted only if its Euclidean distance from the previously plotted point exceeds $\delta_{\min} = 0.3$. Consecutive points are connected by directional arrows and labelled with the interaction index. Coloured background regions delimit the four learning situations defined by the population medians of p_{L_0} and p_T.



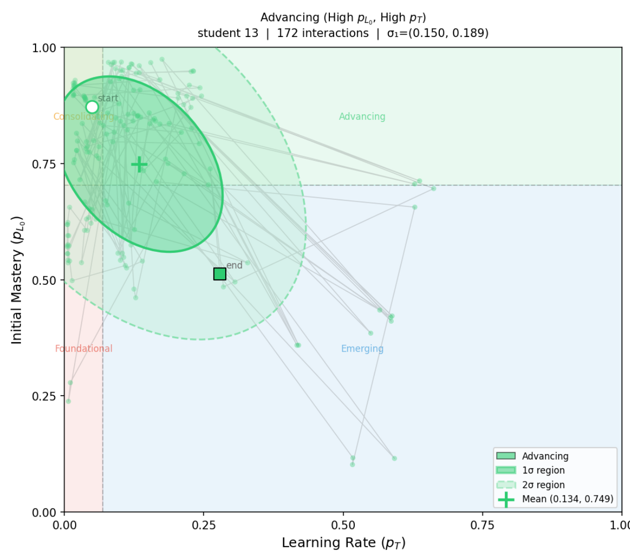
(a) *Foundational* (low p_{L_0}, low p_T): compact ellipse at low mastery and low learning rate; persistent difficulty calls for immediate scaffolding.



(b) *Consolidating* (high p_{L_0}, low p_T): compact ellipse at high mastery and near-zero learning rate; stagnation calls for novel challenge.

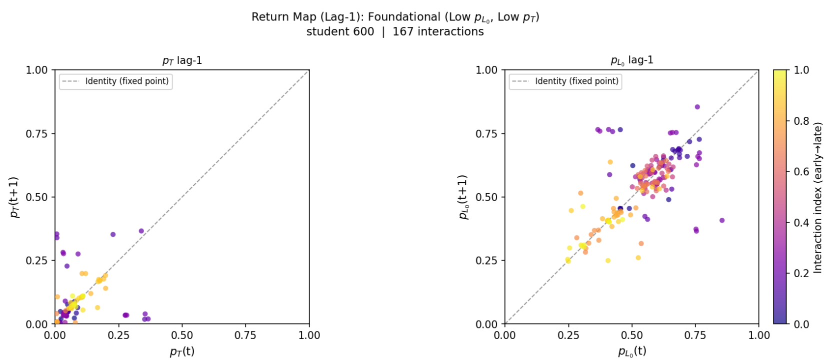


(c) *Emerging* (low p_{L_0}, high p_T): most spatially extended ellipse; heterogeneous profile across skills benefits from skill-level differentiation.

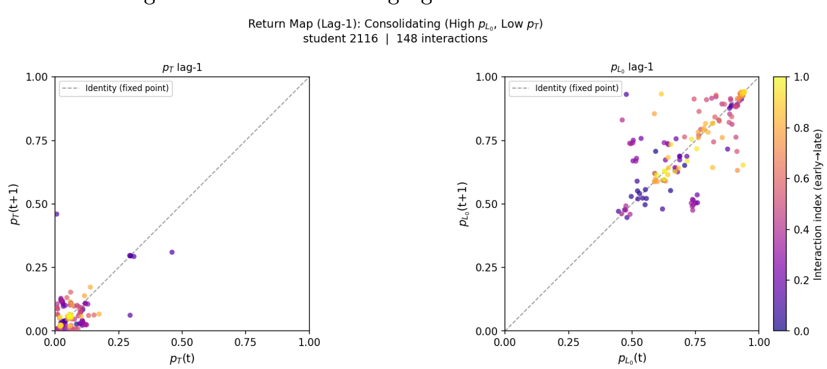


(d) *Advancing* (high p_{L_0}, high p_T): moderately extended ellipse at high mastery and robust learning rate; supports routing to accelerated content.

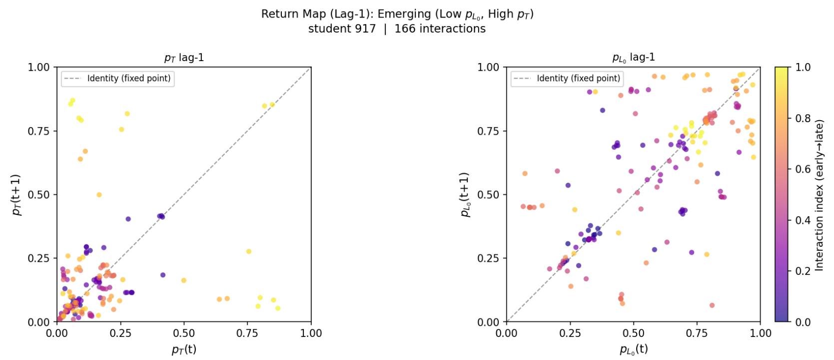
Fig. 3: **Parameter orbit ellipses.** Covariance ellipses (1σ and 2σ) for the representative student of each learning situation in $(p_{\mathrm{L}_0}, p_{\mathrm{T}})$ space. The centre marks the mean parameter position; ellipse shape encodes the extent and orientation of parameter variability across all interactions.



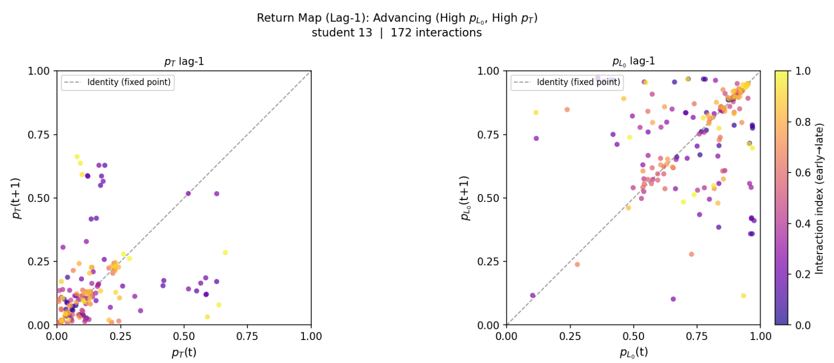
(a) *Foundational* (low p_{L_0}, low p_T): broad scatter with downward mastery drift and a stable low learning-rate attractor converging over time.



(b) *Consolidating* (high p_{L_0}, low p_T): mastery orbit migrates to a high-mastery fixed point while learning rate declines, revealing an active-to-plateau transition.



(c) *Emerging* (low p_{L_0}, high p_T): both mastery and learning rate increase over time, with points distributed above the diagonal in both panels.



(d) *Advancing* (high p_{L_0}, high p_T): mastery clusters at high values from the first interactions; learning rate declines moderately as marginal gains diminish.

Fig. 4: **Return maps.** Plots for the representative student of each learning situation. Each panel shows $p_{\mathrm{L}_0}(t+1)$ vs $p_{\mathrm{L}_0}(t)$ (left) and $p_{\mathrm{T}}(t+1)$ vs $p_{\mathrm{T}}(t)$ (right). Points are coloured from dark (early) to bright (late) interactions. The diagonal (dashed line) marks the fixed point; deviations above indicate increasing parameters and deviations below indicate decreasing ones.

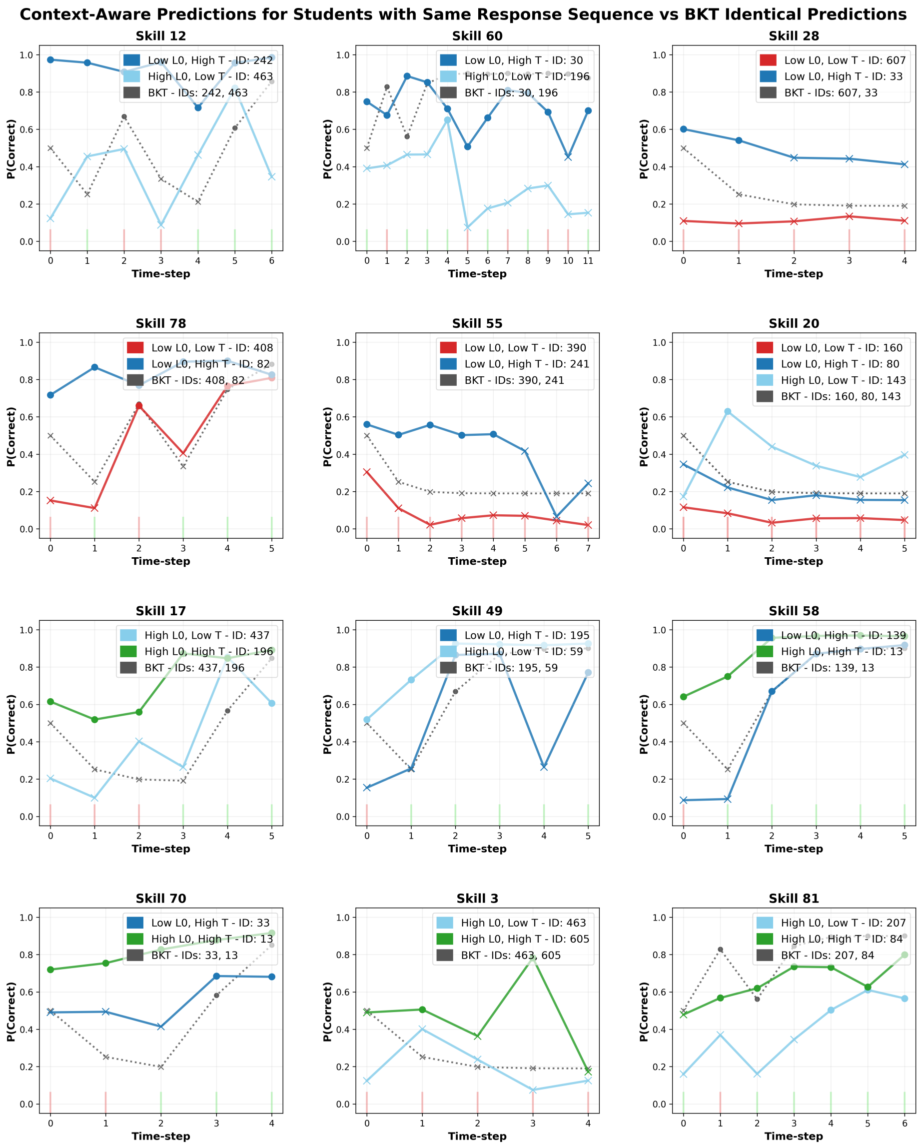


Fig. 5: **Context-aware prediction mosaic.** Each panel shows students with identical response sequences (green/red bars for correct/incorrect responses) but different learning situations. BKT predictions (dotted grey) overlap due to Markovian constraints; gTransformer predictions (solid coloured) diverge based on historical parameters (p_{L_0}, p_{T}). Circles indicate predicted correct responses; crosses indicate predicted failures.